

DEBUGGING AND OPTIMIZATION

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QUESTIONS

Q1

Debug Nested Exception Handling
10 marks

**Q2**

Debug Exception Propagation
10 marks

**Q3**

Debug Multiple Runtime Exceptions
10 marks

**Q4**

Debug Finally Execution
10 marks

**Q5**

Debug the NullPointerException
10 marks



5/5 answered

Q1 Debug Nested Exception Handling 10 marks

```
import java.util.Scanner;

public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = {20,40,60,80};

        int index = sc.nextInt();
        int divisor = sc.nextInt();

        try {
            int value = arr[index];
            try {
                int result = value/divisor;
                System.out.println(result);
            }
            catch(ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Index");
            }
        }
        catch(ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

Debugging Task

Identify why some exceptions are never caught.
Modify the exception handling so all runtime exceptions are handled correctly.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code



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```
10         try {  
11             int result = value / div  
12             System.out.println(resul  
13         }  
14         catch (ArithmeticException e  
15             System.out.println("Divi  
16         }  
17     }  
18     catch (ArrayIndexOutOfBoundsException  
19         System.out.println("Invalid  
20     }  
21     sc.close();  
22 }  
23 }
```

Test Input

2
10

Output

6

Test Cases

Test Case 1



Test Case 2



Test Case 3



6.67 / 10 marks

 Run

 Save

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NullPointerException
10 marks



5/5 answered

Q2 Debug Exception Propagation

10 marks

```
import java.util.Scanner;

class Calculator{

    static int divide(int a,int b){
        return a/b;
    }

    static void display(int x){
        System.out.println("Answer = "+x);
    }

}

public class Debug2{

    public static void main(String args[]){

        Scanner sc=new Scanner(System.in);

        int a=sc.nextInt();
        int b=sc.nextInt();

        Calculator.display(Calculator.divide(a,b));

        System.out.println("Execution Completed");
    }

}

Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
```

TEST CASE 1

INPUT: 20 5

OUTPUT : Answer = 4

Execution Completed

TEST CASE 2

INPUT: 50 0

OUTPUT: Cannot Divide by Zero

Execution Completed

TEST CASE 3

INPUT: 36 6

OUTPUT: Answer = 6

Execution Completed

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Calculator {
3     static int divide(int a, int b) {
4         return a / b;
5     }
6     static void display(int x) {
7         System.out.println("Answer = " +
8     }
9     public static void main(String args[
10         Scanner sc = new Scanner(System.
11         int a = sc.nextInt();
12         int b = sc.nextInt();
13         try {
14             Calculator.display(Calculatc
15         }
16         catch (ArithmeticException e) {
17             System.out.println("Cannot D
18         }
19         finally {
20             System.out.println("Executic
21         }
22     }
```

```
23 }
```

Test Input

```
20 5
```

Output

```
Answer = 4  
Execution Completed
```

Test Cases

Test Case 1



Test Case 2



Test Case 3

**10.00 / 10 marks** **Submitted**



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**Q4**

Debug Finally Execution
10 marks

**Q5**

Debug the
NullPointerException
10 marks



5/5 answered

Q3 Debug Multiple Runtime Exceptions 10 marks

```
import java.util.Scanner;

public class Debug3 {

    public static void main(String args[]) {

        Scanner sc=new Scanner(System.in);

        int arr[]={12,24,36,48};

        String number=sc.next();

        int index=Integer.parseInt(number);

        System.out.println(arr[index]);

    }

}
```

Debugging Task

The program may generate two different runtime exceptions.

Modify the program so that each exception is handled separately.

Test Case 1

INPUT: 2

OUTPUT: 36

Test Case 2

INPUT: ABC

OUTPUT: Invalid Number

Test Case 3

INPUT: 5

OUTPUT: Invalid Array Index

✱ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug3 {
3     public static void main(String args[])
4     {
5         Scanner sc = new Scanner(System.in);
6         int arr[] = {12, 24, 36, 48};
7         String number = sc.next();
8         try {
9             int index = Integer.parseInt(number);
10            try {
11                System.out.println(arr[index]);
12            }
13            catch (ArrayIndexOutOfBoundsException e)
14            {
15                System.out.println("Invalid Array Index");
16            }
17        }
18        catch (NumberFormatException e)
19        {
20            System.out.println("Invalid Input");
21        }
22    }
23 }
```

Test Input

2

Output

36

Test Cases

Test Case 1



Test Case 2



Test Case 3



6.67 / 10 marks

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Exceptions
10 marks

**Q4**

Debug Finally Execution
10 marks

**Q5**

Debug the
NullPointerException
10 marks



5/5 answered

Q4 Debug Finally Execution

10 marks

```
public class Debug4 {  
  
    static int calculate(int a,int b){  
  
        try{  
            return a/b;  
        }  
  
        finally{  
            System.out.println("Resources Released");  
        }  
    }  
  
    public static void main(String args[]){  
  
        System.out.println(calculate(20,0));  
  
    }  
}
```

Debugging Task

Prevent program termination.

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

4

Test Case 2

20 0

Resources Released

Division by Zero

Test Case 3

81 9

Resources Released

9

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 public class Debug4 {
2     static int calculate(int a, int b) {
3         try {
4             return a / b;
5         }
6         finally {
7             System.out.println("Resource
8         }
9     }
10    public static void main(String args[
11        int a = 0;
12        int b = 0;
13        java.util.Scanner sc = new java.
14        a = sc.nextInt();
15        b = sc.nextInt();
16        try {
17            System.out.println(calculate
18        }
19        catch (ArithmeticException e) {
20            System.out.println("Division
21        }
22    }
23 }
```

Test Input

20 5

Output

```
Resources Released  
4
```

Test Cases

Test Case 1



Test Case 2



Test Case 3

**10.00 / 10 marks** **Submitted**

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**Q4**

Debug Finally Execution
10 marks

**Q5**

Debug the NullPointerException
10 marks



5/5 answered

Q5 Debug the NullPointerException

10 marks

```
public class Debug4 {  
    public static void main(String[] args) {  
        String name = null;  
        System.out.println(name.length());  
    }  
}
```

Expected Task

Prevent the program from crashing.

Test Case 1

INPUT: Java

OUTPUT: Length = 4

Test Case 2

INPUT: null

OUTPUT: String is Null

Test Case 3

INPUT: Programming

OUTPUT: Length = 11

 **Debugging Task:** Find and fix the bugs in the code



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```
7         System.out.println("String i  
8     } else {  
9         System.out.println("Length =  
10        }  
11    }  
12 }
```

Test Input

java

Output

Length = 4

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks



Submitted